Current Conservation Does Not Ensure Accurate Equilibrium Geometry: A Source-Only Reconstruction Study

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Abstract

Predicting a tokamak's poloidal flux through a current-space Green operator offers a natural way to impose total-current constraints. Whether those constraints improve equilibrium geometry is a separate empirical question. We examine this distinction on a discharge-separated DIII-D development cohort with coil currents, plasma current and Thomson profiles, without magnetic probes or flux loops. A fixed 32-column current dictionary supports a convex correction that minimizes flux displacement subject to nonnegative cell currents and the observed current sum. All 256 predictions pass the declared numerical feasibility and optimality checks. The median displacement is 0.451498 times that of a simpler clipped correction, yet its local composite reconstruction score is 0.783302, versus 0.869123 for direct flux regression, and its boundary distance increases from 0.041442 to 0.060279. A separate stored-map norm bound excludes removing half the recorded error gap merely by replacing one fixed compressed current estimator with its full-output counterpart. These results distinguish representation, numerical optimization and predictive utility. They do not establish a limitation of all current-space methods, an original reconstruction mechanism, or transfer to MAST. Preserved protocols, failed solver attempts, arrays and independent readbacks make the negative results inspectable.

1 Question and relationship to established methods

The Fusion Equilibrium Challenge asks for equilibrium flux and two scalar quantities from active-coil currents, total plasma current, electron Thomson scattering and available geometry [1, 2]. Its source and target machines differ substantially. A physically meaningful representation is useful, but cannot by itself supply information missing from the observations. The present question is deliberately narrower than a zero-shot inverse solution: *does enforcing a nonnegative, current-conserving source with minimum flux displacement improve the geometry of a fixed source-trained predictor?*

Current-space reconstruction with Green responses is established. McClenaghan et al. construct current principal components and corresponding flux and diagnostic bases, and reuse source-device components with a target-device Green operator [3]. Their magnetic observations are outside the restricted input contract tested here; no matched reproduction is claimed. BEAST separately assesses consistency between Biot–Savart current models and Grad–Shafranov force balance [4]. Thus current conservation, magnetostatic decoding and force-balance compatibility should not be conflated. The current dictionary, kernel regression and convex projection used below are conventional controls, not claimed inventions.

Our evidence is a local, repeatedly reused development cohort. It supports a failure of one fixed intervention, rather than a general impossibility result or a new leading benchmark. Earlier contour-relabeling and synthetic ambiguity notes remain separate artifacts; they are not merged into a claim that an original inverse method has been found.

2 Physical convention and supervised representation

For per-radian flux ψ in Wb/rad and coordinates in metres,

$$B_R = -\frac{\psi_Z}{R}, \qquad B_Z = \frac{\psi_R}{R}, \qquad \Delta^*\psi = \psi_{RR} - \frac{\psi_R}{R} + \psi_{ZZ} = -\mu_0 R j_\phi. \tag{1}$$

Static, axisymmetric, isotropic-pressure force balance additionally requires

$$j_\phi = R p'(\psi) + \frac{F(\psi)F'(\psi)}{\mu_0 R}. \tag{2}$$

A Green-decoded arbitrary current need not satisfy Eq. (2). Electron profiles alone do not specify total pressure, which may include ion and fast-particle contributions. No force-balance certificate is inferred from the benchmark's map-derived geometric functionals.

We use the retained label convention $\chi = -y$, where y is the released flux raster, and an offline shot orientation $\sigma \in \{-1, 1\}$ from the dominant measured plasma-current sign. All 21 current features are multiplied by this orientation. Under this convention the corresponding physical field would have flux $\sigma\chi$; this does not independently verify upstream sign or calibration provenance. The orientation uses the full pulse and is not a causal real-time input.

On the 65×65 grid, the supervised source construction differentiates the full reference χ raster using bicubic interpolation, then masks the result by its extracted plasma support. It clips negative canonical current and normalizes the positive cell masses to $I = |I_p|$. For cell area A_i,

$$m_i = I\rho_i, \qquad \rho_i \geq 0, \quad \mathbf{1}^\mathsf{T}\rho = 1, \qquad \widehat{\chi} = Gm + f + h. \tag{3}$$

The last expression is a representation ansatz, not an exact identity for the reference raster. Here G maps amperes of cell current to Wb/rad, f is the modeled known F-coil field and h is a six-column exterior/gauge expansion. Current uses $\mathrm{d}R\,\mathrm{d}Z$, without a toroidal-volume factor $2\pi R$. The current guards concern canonical magnitude I. At the retained weak-sign development frame (row 1504), σI differs from the signed instantaneous input current; signed physical-current agreement is not claimed there. The decoder uses integrated finite-cell self responses and order-eight finite-conductor quadrature. The exterior dictionary contains a constant, one E-coil response and four vacuum-harmonic polynomials. Its training coefficients are label-fitted; at prediction time a fixed kernel regressor supplies them. No validation support or exterior coefficient is an input to the predictor. Unknown E-coil calibration and passive fields remain limitations.

The training mean of ρ and 31 principal components form $B \in \mathbb{R}^{4225\times 32}$. The mean column has unit sum, so it provides a feasible current shape. Let $a \in \mathbb{R}^{32}$ denote absolute-current coordinates, $H = GB$ and $s = B^\mathsf{T}\mathbf{1}$. A trained regressor uses targets $a_{\mathrm{train}} = I_{\mathrm{train}}[1; c_{\mathrm{train}}]$, where c_{train} are the 31 current principal-component coordinates, with training-only target standardization. It predicts d; the unadjusted source is Bd, its total is $s^\mathsf{T}d$, and its flux contribution is Hd. These columns include a mean, whereas the direct-flux comparator uses 32 principal components. Equal column count does not make the representations identical.

3 Fixed conservation controls and attribution bound

3.1 Affine and nonnegative corrections

The unadjusted absolute-coordinate predictor is denoted A. A simpler control sets the mean coefficient to I, conserves the resulting sum, then clips negative cell masses and renormalizes; this is C. Clipping may leave the dictionary span. Neither C nor A is a Grad–Shafranov solver.

To isolate the displacement required by conservation in the same dictionary, consider

$$\min_a \ \frac{1}{2}\,\|H(a-d)\|_2^2\,, \qquad s^\mathsf{T}a = I. \tag{4}$$

When H has full column rank and $s \neq 0$, the unique solution is

$$a_{\mathrm{aff}} = d + \frac{(H^{\mathsf{T}}H)^{-1}s}{s^{\mathsf{T}}(H^{\mathsf{T}}H)^{-1}s}(I - s^{\mathsf{T}}d). \tag{5}$$

This follows from stationarity of the equality-constrained quadratic and substitution of its multiplier into the equality. The implementation uses an SVD rather than explicitly forming the displayed inverse. The retained H has rank 32 and condition number approximately 301. Every affine-corrected case violates the chosen nonnegative-current family. This is not a claim that all physical plasmas have nonnegative canonical current everywhere.

The primary positive-current control therefore solves, with $z = d/I$,

$$\min_{u\in\mathbb{R}^{32}} \frac{1}{2}\left\|M(u-z)\right\|_2^2, \qquad s^{\mathsf{T}}u = 1, \quad Bu \geq 0, \qquad M = H/\sigma_{\max}(H). \tag{6}$$

All 4225 original inequalities are retained. The mean column supplies a feasible point. Full column rank makes the objective strictly convex, hence the feasible minimizer unique. These elementary finite-dimensional facts do not certify the accuracy of the learned source, its plasma support, its field topology, or Eq. (2). The corrected label prediction is $\widehat{y} = -(IHu + f + \widehat{h})$, with the same known field f and predicted exterior $\widehat{h}$ as A.

The implementation uses Clarabel 0.11.1 [5] with direct QDLDL factorization, fresh solves, and declared 10^{-12} feasibility/gap tolerances. External checks on original arrays require relative current error and negative-mass fraction at most 10^{-10}, normalized stationarity at most 10^{-7}, dual-sign violation at most 10^{-10} and complementarity at most 10^{-9}. They also check finite maps, the training-support union and an objective no worse than the feasible mean. Numerical admission is toleranced, rather than a computer-assisted proof of exact real-arithmetic feasibility. Two earlier solver recipes, SLSQP and OSQP, stopped unresolved under the same external requirements. Their partial arrays and failed admission remain preserved; successful numerical completion did not erase them.

3.2 A stored-map comparison independent of new labels

An apparent current-regression deficit might instead arise from compression. One fixed test replaces the normalized-current PCA target by a full 4225-output centered kernel target, with identical kernel, inputs and readouts. This is supervised kernel regression, not a fit-free operation. For training targets Y with mean μ and fixed kernel weights W, its output is $\mu + W(Y - \mu)$, followed by the unchanged signed or clipped normalization. No new validation reference is needed for the following comparison.

Proposition 1 (Classical norm bound for the fixed stored maps)**.** *For discharge q, let P_q, Q_q, T_q be two prediction arrays and any common reference. Let $\|\cdot\|_q$ be the RMS norm over all retained frames and grid cells of that discharge. Define $E(P,T) = n^{-1}\sum_q \|P_q - T_q\|_q$ and $D(P,Q) = n^{-1}\sum_q \|P_q - Q_q\|_q$. Then*

$$|E(P,T) - E(Q,T)| \leq D(P,Q). \tag{7}$$

Proof. The reverse triangle inequality bounds each discharge's difference by $\|P_q - Q_q\|_q$. Sum those bounds and divide by n. □

This result applies to any common reference. The subsequent comparison to a *recorded* performance gap uses only the original cohort's particular gap, not the possibly different gap for another reference. Upward-rounded norm arithmetic guards the stored binary64 arrays under standard floating-point assumptions; it is not an interval enclosure of a continuum PDE solution.

4 Development experiment and review boundary

The dataset is pinned to revision `1e280905b85f2a6fdde7e06fca8cf3a1edf447cb`; scorer 3.2.0 is pinned to starter commit `823f6eef1f527e1b5566a8fc1e952da6e2fadf43` [2, 6]. The pilot has 96 training discharges and 16 development discharges, each with 16 approximately evenly spaced retained EFIT frames: 1536 training and 256 development examples. A historical 16-discharge test partition is excluded from the new studies. Discharge separation prevents adjacent frames of the same pulse appearing in both fit and development sets. Repeated development reuse still prevents treating these results as fresh confirmation or official challenge eligibility.

The basic 121 raw features contain currents and Thomson summaries/bins; training-only imputation, missingness indicators and scaling produce 175 columns. The RBF kernel uses $\alpha = 0.1$, $\gamma = 1/175$. Direct flux regression uses 32 flux components. Current and exterior regressors use the same inputs/kernel. Scalar predictions q_{95}, β_N are bitwise identical across A, C, the positive QP and the direct RBF comparison. Additional fixed controls use a four-layer, 512-unit ReLU network trained for 500 epochs, and 273 raw temporal/physical-coordinate features (378 after preprocessing). Their retained fixed runs provide context, without selecting a new winner. The neural run ended without convergence; richer temporal features include future samples, and none of these runs demonstrates causal control.

The local score is

$$S = 0.55[R^2_\psi]_+ + 0.15[R^2_{qb}]_+ + 0.10(1 - \min(1, D_{\mathrm{LCFS}})) + 0.20C, \tag{8}$$

where C averages nonnegative parts of seven map-functional R^2 values. Flux/scalar R^2 is pooled; the separately reported raw-gauge flux RMSE averages discharge RMS norms equally. D_{LCFS} is the pinned scorer's normalized boundary distance. Its geometric consistency term is neither a current integral nor a force-balance residual. The local wrapper chooses one global sign using all scored pixels; the official parallel evaluator instead describes a subsample probe. No per-frame signs are chosen. The direct and QP maps are both scored with +1, and their reported probe diagnostic agrees with this unflipped orientation. These local results are not official scores.

The input-only projection study was frozen and completed before a separate fixed performance protocol opened development references. The primary gate requires a score greater than direct RBF, flux RMSE at most 95% of its value, and no greater LCFS distance, retaining all 256 finite predictions. Signed affine correction is a declared secondary contrast and cannot rescue failure of the nonnegative primary. Exactly two paired RMSE intervals resample whole discharges 2000 times with seed 20261029. They are descriptive under repeated development reuse. Independent reviewers checked frozen hashes, saved predictions, currents/KKT residuals, metric arithmetic and these intervals. They did not independently repeat the contour and seven-functional extraction.

5 Results: feasible sources, degraded geometry

All 256 positive QPs pass numerical admission. The median per-frame RMS displacement from A is 0.002530 Wb/rad, versus 0.005604 for C, a ratio of 0.451498. This displacement premise passes; it is not the performance gate.

The positive QP scores 0.783302 and has RMSE 0.041840, exceeding the frozen threshold 0.039484. Its boundary distance 0.060279 exceeds direct RBF's 0.041442. It therefore fails all three performance requirements despite passing every current/optimization guard. Flux R^2 rises slightly from 0.934024 to 0.935819, while the geometric consistency term falls from 0.656065 to 0.231442. The QP's untruncated axis-R, axis-Z, elongation and volume R^2 values are negative. The score clips those negative values to zero before averaging; reporting only the positive consistency average would hide their severity. One of 256 QP contours fails extraction, versus none for direct RBF; no reference frame is dropped. Figure 1 separates these metrics.

Method	Inputs	$S \uparrow$	RMSE$\downarrow$	$D_{\mathrm{LCFS}} \downarrow$
Matched 175-input current comparison				
Direct RBF	175	0.869123	0.041562	0.041442
A: unconstrained	175	0.846165	0.034978	0.052167
C: clipped	175	0.802864	0.046544	0.049441
Positive-current QP	175	0.783302	0.041840	0.060279
Additional baselines, same cached scalar predictions				
MLP	175	0.877589	0.044062	0.043400
Rich-input RBF	378	0.885070	0.040087	0.039086

Table 1: Retained local development results. All rows use the same cached scalar predictions; "richer" changes available input features. RMSE is in Wb/rad; D_{LCFS} is dimensionless. These are conventional baselines and fixed failed interventions, not independent blind-test results.

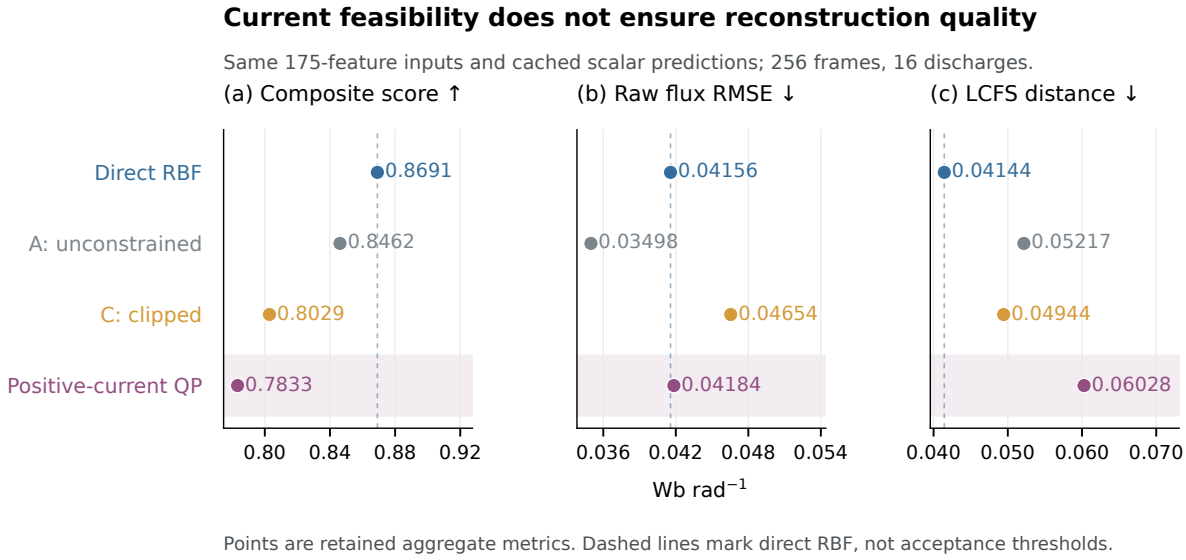


Figure 1: Numerical constraints and predictive utility answer different questions. All plotted data are read from the saved evaluations; no scoring or fitting is performed to make this figure. Smaller raw flux error does not imply better boundary or map-functional recovery. The same scalar predictions are used throughout.

The signed affine contrast attains lower RMSE, 0.0384794 Wb/rad, but lower score 0.789399 and worse boundary distance 0.071311, while violating the chosen nonnegative family on all cases. This prevents interpreting raw flux error as a sufficient objective. The whole-discharge mean RMSE differences and descriptive 95% intervals, in Wb/rad, are

$$\begin{aligned} \text{A minus QP:} \quad & -0.006862, \quad [-0.014027, \ -0.001048], \\ \text{Direct minus QP:} \quad & -0.000278, \quad [-0.003677, \ 0.003820]. \end{aligned}$$

The latter does not establish a nonzero mean flux-error difference. Its uncertainty does not undo the observed failure of the fixed multi-metric gate. Figure 2 displays the previously declared worst QP flux-error frame, not a representative randomly sampled case.

The separate compression test gives conservative stored-map distances of 2.40349×10^{-4} and 2.37055×10^{-4} Wb/rad for the signed and clipped readouts. Each is less than half its own recorded current-minus-direct RMSE gap; the half-gap thresholds are approximately 2.42×10^{-3} Wb/rad. By Eq. (7), replacing that particular compressed estimator by its fixed full-output counterpart cannot close half that gap. This result does not exclude other kernels, capacities, representations or physical methods, and no newly scored full-output accuracy is implied.

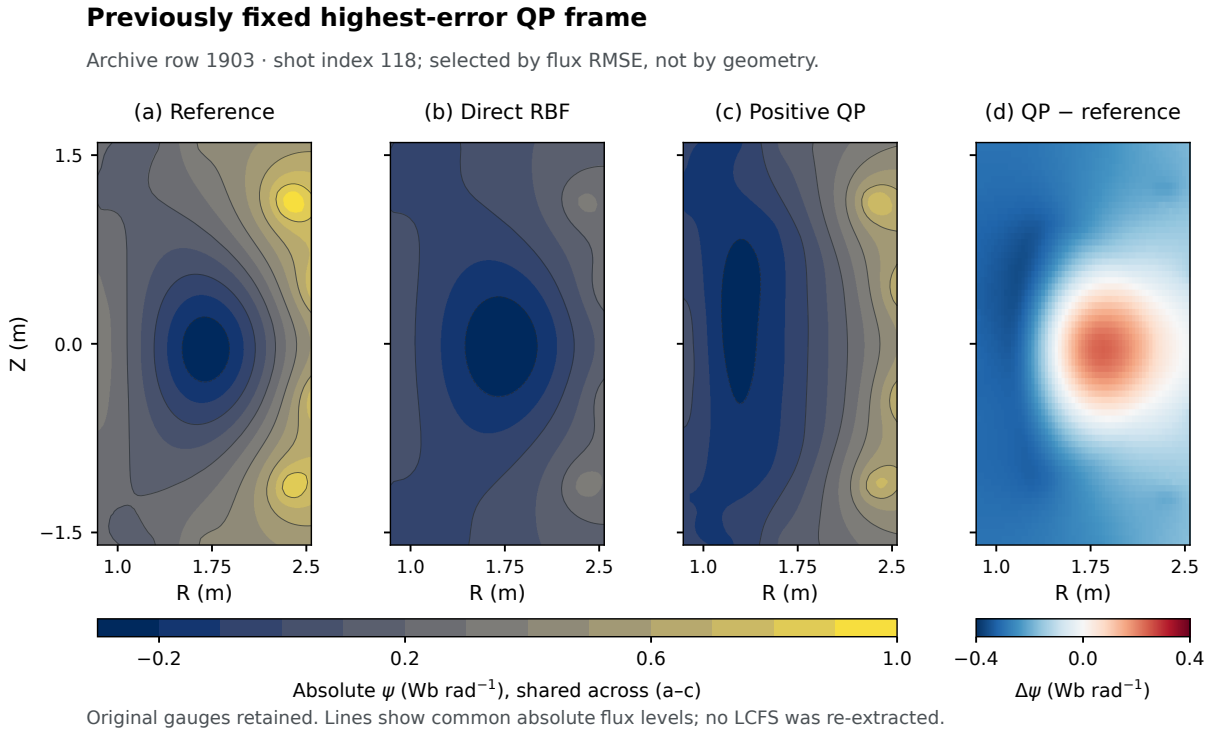


Figure 2: Descriptive failure example, development row 1903, shot index 118, selected by the fixed evaluation's highest QP raw per-frame RMSE. Physical R, Z coordinates retain equal aspect ratio. Reference, direct and QP maps use the released label convention and shared absolute flux levels; no individual gauge, sign or amplitude alignment is fitted. Display contours are visual isolines, not a second independent LCFS or force-balance evaluation.

6 Interpretation, source accounting and limitations

The supported conclusion is conditional: for these fixed learned coordinates, dictionary, nuisance field and development shots, minimum-flux displacement under nonnegative current conservation did not yield useful equilibrium geometry. Neither the finite-dimensional optimum nor a small field norm controls critical-point or separatrix recovery without further assumptions. This is a motivation for explicit geometric and force-balance evaluation, not evidence that harder constraints always help or hurt.

The source teachers themselves require scrutiny. Thirteen normalization gains fall below 0.5: eleven training cases and two development cases, all retained. A training-only derivative audit tests whether the sampled known F-coil field can explain the eleven training failures. With fixed support and area, let $j_0 = -M_\Omega \Delta_h^* \chi/(\mu_0 R)$, $f_h = -M_\Omega \Delta_h^* f/(\mu_0 R)$ and $j_1 = j_0 - f_h$, where M_Ω is the unchanged binary support mask. Since positive part is 1-Lipschitz,

$$|A\sum_i [j_{0,i}]_+ - A\sum_i [j_{1,i}]_+| \le A\sum_i |f_{h,i}|. \tag{9}$$

The retained bound is too small to restore any of those eleven cases to the predeclared gain interval. This rejects the modeled-F explanation in that discrete scope; it does not certify the differentiated reference source. All eleven are first retained EFIT frames. Input-only forensic readback matches their small native currents exactly at stored precision and confirms that the pinned timing correction was applied. It does not identify an upstream clock, equilibrium initialization or reference-quality defect. Unknown passive/E-coil fields, source support and raster differentiation remain distinct possibilities. No frames are removed and no current or clock is calibrated against labels.

No experimental MAST reconstruction data, weights or derived statistics are used for training, tuning, calibration, validation or selection. During prior literature inspection one printed target-profile numerical example was incidentally displayed; its numbers were never copied into priors, code, data or model decisions. No MAST accuracy, official hidden-label score, fresh independent reproduction, deployment or real-time operation is claimed. Observed runtime is descriptive: the 256 completed QPs took about 30.5 s in one process with peak RSS about 506 MB. QDLDL reports one thread, but other loaded numerical pools and Apple Accelerate parallelism are not uniformly controlled. These measurements do not support a fair hardware-speed ranking.

7 Reproduction and remaining scientific requirement

The task-local `REPRODUCE.md` records pinned source/data hashes, protocols, numerical dependencies, solver wheel licenses and the sequence of saved-array checks. The dense 4225^2 binary64 Green matrix occupies about 143 MB; the reduced response has 32 columns. Dense Green construction/storage scale quadratically in grid-cell count; a reduced prediction costs $O(Nk)$ for N cells and $k = 32$ coordinates, apart from kernel evaluation and solver work. No improved asymptotic complexity is claimed. Preserved negative and unresolved outcomes are part of the evidence, rather than omitted trials. Figure/table scripts regenerate these displays from retained arrays without retraining or re-evaluating predictions.

Further work requires an inverse mechanism beyond the tested representations, improvements over matched baselines on fresh discharges, and permitted cross-machine evaluation. The present results identify the limits of the retained current-constrained reconstruction method.

References

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